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FORM TP 2006264

MAY/JUNE 2006

CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

PHYSICS

UNIT 01 – Paper 02

2 hours 15 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of **NINE** questions.
2. Section A consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in this answer booklet.
3. Section B consists of **SIX** questions. Candidates must attempt **THREE** questions in this section, **ONE** question from **EACH** Module. Answers for this section must be written in the separate answer booklet provided.
4. All working **MUST** be **CLEARLY** shown.
5. The use of non-programmable calculators is permitted.

NOTHING HAS BEEN OMITTED

LIST OF PHYSICAL CONSTANTS

Universal gravitational constant	G	=	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Acceleration due to gravity	g	=	9.80 m s^{-2}
Radius of the Earth	R_E	=	6380 km
Mass of the Earth	M_E	=	$5.98 \times 10^{24} \text{ kg}$
Mass of the Moon	M_M	=	$7.35 \times 10^{22} \text{ kg}$
1 Atmosphere	Atm	=	$1.00 \times 10^5 \text{ N m}^{-2}$
Boltzmann's constant	k	=	$1.38 \times 10^{-23} \text{ J K}^{-1}$
Density of water		=	$1.00 \times 10^3 \text{ kg m}^{-3}$
Thermal conductivity of copper		=	$400 \text{ W m}^{-1} \text{ K}^{-1}$
Specific heat capacity of aluminium		=	$910 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific heat capacity of copper		=	$387 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific heat capacity of water		=	$4200 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific latent heat of fusion of ice		=	$3.34 \times 10^5 \text{ J kg}^{-1}$
Specific latent heat of vaporization of water		=	$2.26 \times 10^6 \text{ J kg}^{-1}$
Avogadro's constant	N_A	=	6.02×10^{23} per mole
Molar gas constant	R	=	$8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Stefan-Boltzmann's constant	σ	=	$5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Speed of light in vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$

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SECTION A

Attempt ALL questions. You MUST write in this answer booklet.

1. The data in Table 1 below were collected at a laboratory which investigates damage to cars in collisions. In this case a car with a mass of 1200 kg was driven into a solid wall.

Table 1

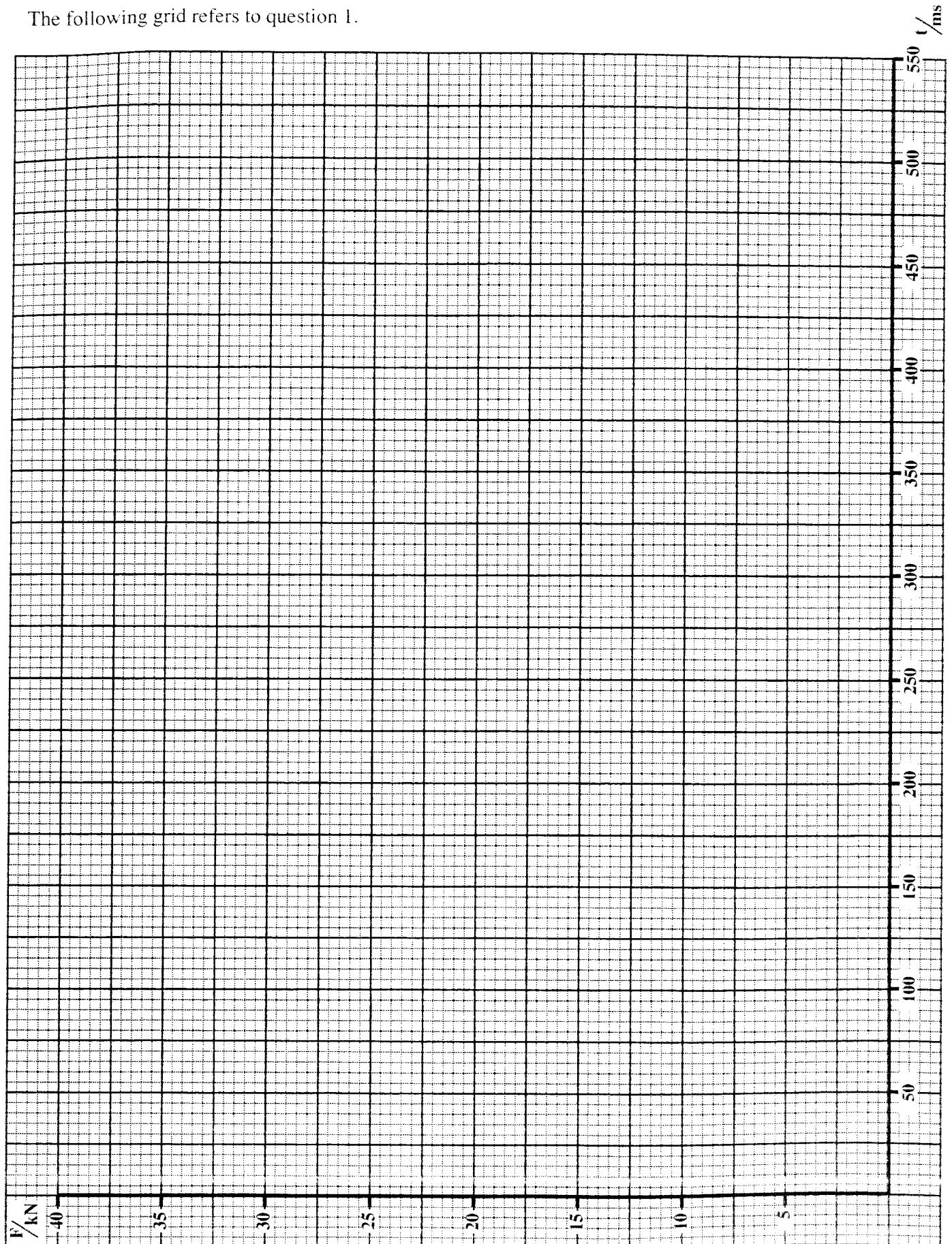
t/ms	F/kN
0	0
50	6.5
100	16.5
150	26
200	32.5
250	37.
275	37.5
300	37
350	33
400	26
450	14
500	6
550	1
575	0

- (a) Use the data to plot a graph showing how the force on the car varies with time.

[3 marks]

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The following grid refers to question 1.



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- (b) (i) What physical quantity is represented by the area under the graph?

- (ii) If the area under the graph is 186 one-centimetre squares what is the value of this quantity using the units given?

[3 marks]

- (c) Find the speed of the car just before impact, assuming it came to rest in contact with the wall.

[2 marks]

- (d) What was the maximum acceleration experienced by the car?

[2 marks]

Total 10 marks

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NOTHING HAS BEEN OMITTED

2. Figure 1 shows an experimental arrangement set up to measure the focal length, f , and position, d , of a converging lens mounted in an inaccessible position inside a cardboard tube AB with windows at both ends. The distance from end A to the screen is x .

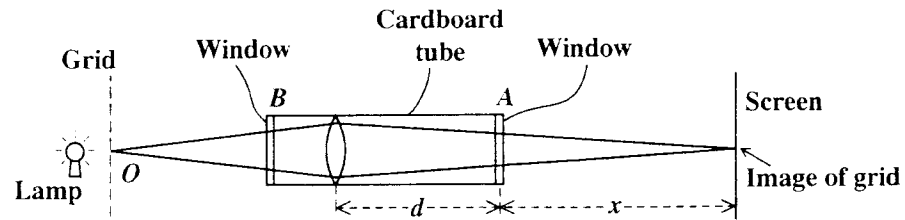


Figure 1

- (a) Briefly explain how you would take readings in order to determine the magnification of the image for various values of x .

[3 marks]

- (b) The magnification, m , of the image, is given by $m = \frac{(x + d)}{f} - 1$.

Figure 2 is a plot of m versus x .

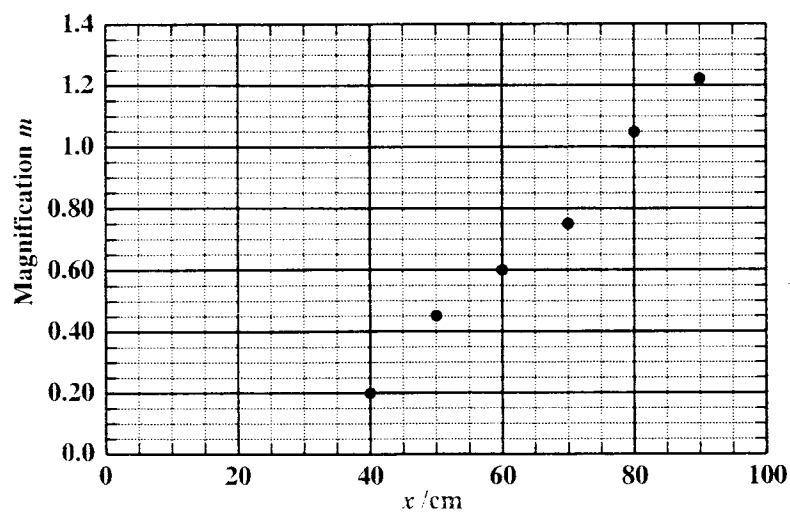


Figure 2

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Show that:

- (i) the gradient of the graph is $\frac{1}{f}$

[1 mark]

- (ii) when $m = 0$, $x + d = f$.

[1 mark]

- (c) (i) Draw the best straight line through the points.

[1 mark]

- (ii) Determine

- a) the gradient of the graph

[2 marks]

- b) the focal length, f , of the lens

[1 mark]

- c) the position d of the lens inside the tube.

[1 mark]

Total 10 marks

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3. A thermistor is to be calibrated by plotting a graph showing its resistance at various temperatures. The table below shows the data collected.

- (a) (i) Draw a circuit diagram showing how the values of the p.d. across the thermistor and the current flowing through it might be measured. [1 mark]

- (ii) Describe briefly the experimental set-up and suggest ONE precaution which might be taken to ensure accurate values.

[2 marks]

- (b) (i) Complete Table 2 to show the values of the resistance, R .

Table 2

$\theta / ^\circ\text{C}$	p.d./V	I/mA	R/Ω
0	2.04	2.74	
10.0	2.03	3.50	
20.0	2.02	5.05	
30.0	1.99	6.75	
40.0	1.97	8.90	
50.0	1.95	11.9	
60.0	1.95	15.9	
70.0	1.95	19.9	
80.0	2.02	28.0	

[2 marks]

- (ii) On the page opposite plot a graph of resistance, R , versus temperature, θ .

[4 marks]

- (c) Find the temperature indicated by this thermometer when the resistance is $320\ \Omega$.

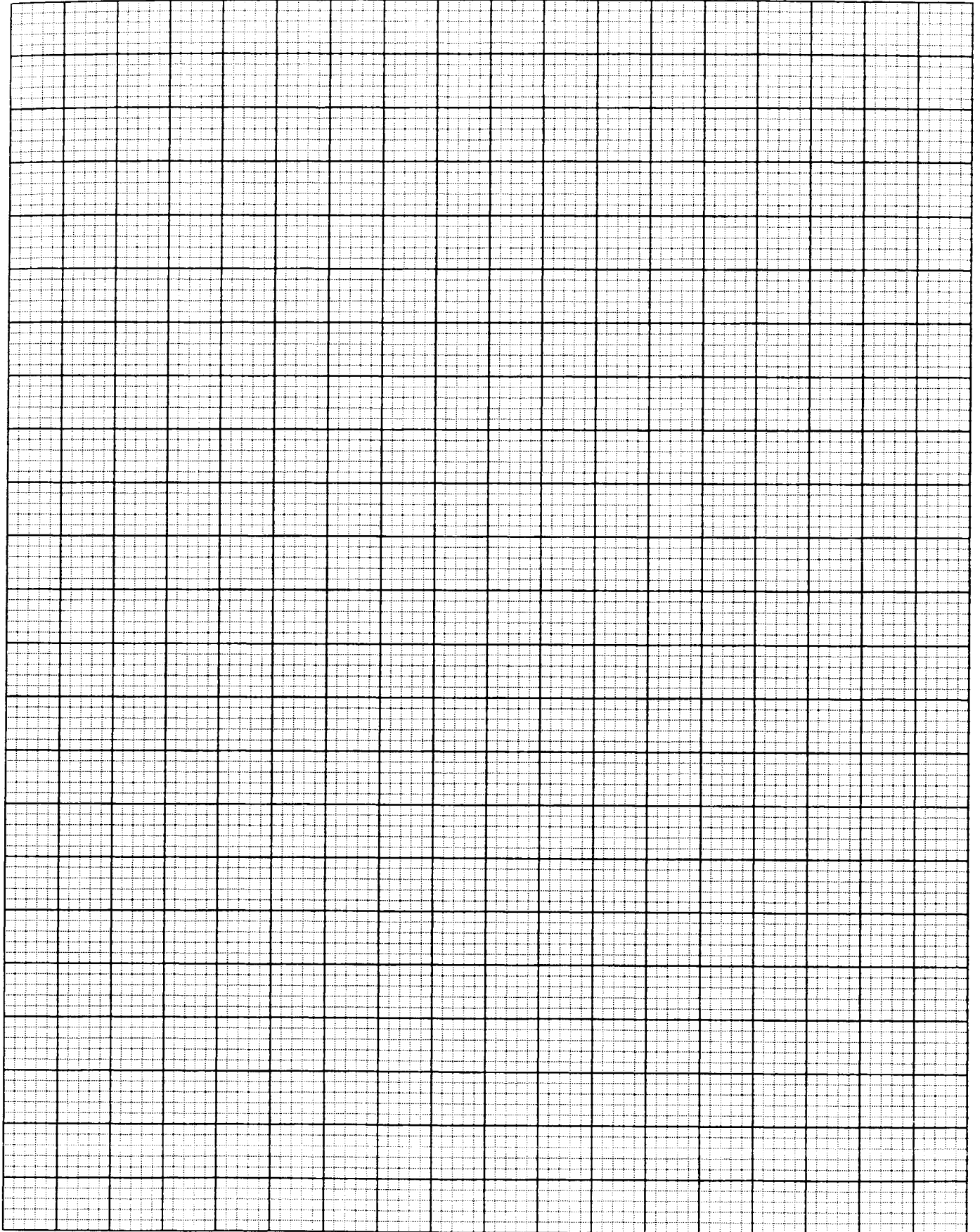
[1 mark]

$$\theta = \text{ } ^\circ\text{C}$$

Total 10 marks

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The following grid refers to question 3(b).



SECTION B

You must attempt **THREE** questions from this section. Choose **ONE** question **EACH** from Module 1, 2 and 3. You **MUST** write your answers in the separate answer booklet provided.

MODULE 1

Answer **EITHER** Question 4 **OR** Question 5.

4. (a) (i) State Newton's laws of motion.
- (ii) Explain why a small sphere, released from rest at the surface to fall through liquid in a deep tank, eventually moves with a constant speed. Sketch a graph showing how the acceleration of the sphere varies with time after its release at the surface of the liquid. **[8 marks]**
- (b) A large helium weather balloon is spherical in shape with a radius of 2.5 m and a total mass of 15 kg (balloon plus helium plus equipment). The balloon is released from sea level and travels vertically upward, quickly reaching its terminal velocity.

The drag force acting on the balloon is given by

$F_D = \frac{1}{2} \pi r^2 \rho v^2$, where r is the radius of the balloon, ρ is the density of air, and v is the ascension speed of the balloon.

- (i) Draw a free body diagram to show the forces acting on the balloon.
- (ii) a) Use Archimedes' principle to find the buoyant force acting on the balloon.
- b) Calculate the terminal velocity of the balloon.
- c) Find the time, in minutes, it will take for the balloon to ascend to a height of 10 km.

(Density of air, $\rho = 1.29 \text{ kg m}^{-3}$.)

(Volume of a sphere = $\frac{4}{3} \pi r^3$.)

[12 marks]

Total 20 marks

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5. (a) A mass, m , moves with constant linear speed, v , in a circle of radius r . Show that the magnitude of the acceleration, a , of the mass is given by

$$a = \frac{v^2}{r} . \quad [6 \text{ marks}]$$

- (b) (i) A steel ball of mass m is attached to an inextensible string of length l . The ball is whirled in a vertical circle. Draw diagrams showing the forces acting on the ball when it is (a) at the top and (b) at the bottom of the circle.
- (ii) The speed of the ball is slowly increased. The string breaks if the tension in it exceeds a critical value, $T_{\max}$. Show that when this critical value is reached the angular speed of the ball is given by

$$\omega = \sqrt{\frac{T_{\max} - mg}{ml}}$$

- (iii) The mass of the ball is 0.5 kg and the string breaks when the angular speed of the ball is 4.0 rad s^{-1} . The length of the string is 1.00 m. Calculate the critical value, $T_{\max}$. [7 marks]

- (c) (i) With the aid of a sketch, describe the motion of the ball after the string breaks at the lowest point of the circle.
- (ii) After the string breaks, it takes 0.5 s for the ball to hit the ground.

Calculate

- a) the lateral distance the ball travelled during this time
- b) the vertical velocity of the ball just before it strikes the ground.

[7 marks]

Total 20 marks

MODULE 2

Answer EITHER Question 6 OR Question 7.

6. (a) Explain the following terms as they relate to the human eye:
- (i) Accommodation
 - (ii) Astigmatism
 - (iii) Cataract
- [3 marks]
- (b) Using lenses with a power of 2.5 D in her glasses, a woman can comfortably read a book held 25 cm from her eyes.
- (i) With the aid of a diagram explain the eye defect from which she suffers.
 - (ii) Determine the woman's near point (i.e. the distance from her eyes where she can see objects clearly without her glasses).
- [7 marks]
- (c) To answer this part of the question you may need to refer to the graph below.

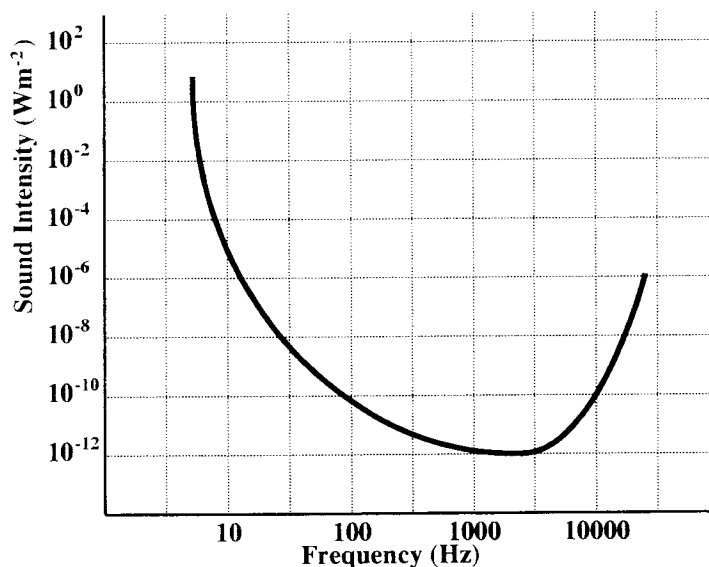


Figure 3: Response curve for the human ear

- (i) What intensity of sound is considered to be the threshold of human hearing?
 - (ii) Why is the response of the human ear said to be logarithmic?
 - (iii) Explain why sound intensity levels are often measured using a decibel (dB) scale.
 - (iv) A noisy motorbike emits noise with an intensity of $6 \times 10^{-2} \text{ W m}^{-2}$. What is the intensity level in dB?
 - (v) Over what range of frequencies would a normal human ear be able to hear a sound with an intensity level of 20 dB?
- [10 marks]

Total 20 marks

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7. (a) (i) Figure 4 shows an undamped mass-spring system. With the aid of sketch graphs, give an account of the energy transformations which occur during one complete cycle of this system.

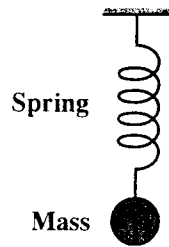


Figure 4

- (ii) Explain why a damped oscillating system experiences a progressive decrease in its amplitude

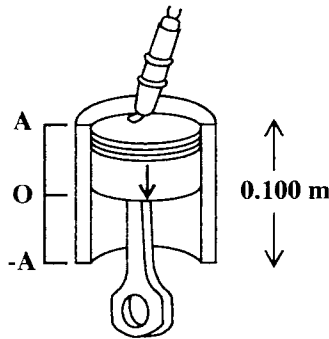


Figure 5

[8 marks]

- (b) Figure 5 shows the piston of an automobile engine. The motion of the piston is simple harmonic. The stroke of the engine (twice the amplitude) is 0.100 m and the engine runs at 3500 revolutions per minute. The mass of the piston is 0.450 kg.

Calculate

- (i) the acceleration of the piston at the endpoint of the stroke
- (ii) the speed of the piston at the midpoint of the stroke
- (iii) the kinetic energy of the piston at the midpoint of the stroke
- (iv) the average power required to accelerate the piston from rest to the speed found in (b) (ii).

[12 marks]

Total 20 marks

MODULE 3

Answer EITHER Question 8 OR Question 9.

8. (a) (i) State TWO mechanisms which enable thermal conduction to take place in a metal.
- (ii) Explain why only one of the mechanisms is responsible for thermal conduction in an insulator.
- (iii) The thermal conductivity, k , of a material may be defined according to the expression

$$\frac{Q}{t} = kA \frac{\Delta\theta}{\Delta x}$$

State the conditions that must be satisfied before the expression may be applied.

- (iv) Explain what is meant by the specific latent heat of vaporization of a substance.
[8 marks]

- (b) Figure 6 shows a stainless steel pot with a flat base 8.50 mm thick that rests on a hot plate. The area of the bottom of the pot is 0.150 m². The water inside the pot is at 100° C, and 0.450 kg of water is evaporated every 3 minutes.

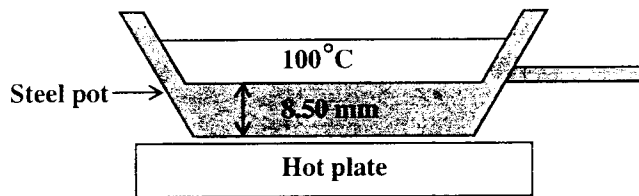


Figure 6

Calculate the surface temperature of the hot plate, which is in contact with the pot.
(Coefficient of thermal conductivity, k , for stainless steel = 50.2 W m⁻¹K⁻¹)

[5 marks]

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- (c) The rate, P , at which an object radiates energy, is proportional to the fourth power of the absolute temperature, T . The intensity of the radiation a distance, r , from the source is given by $I = \frac{P}{4\pi r^2}$.

These relationships may be investigated by using a thermopile connected to a micro-voltmeter and exposing it to the radiation from a small heated sphere that radiates as a point source.

Table 3 shows the micro-voltmeter readings, V , obtained when the sphere was maintained at various temperatures and the thermopile was a distance r from it. The voltage, V , of the thermopile is proportional to the intensity of radiation falling on it.

Table 3

r/m	T/K	$V/\mu\text{V}$
1.5	1500	4.5
1.5	2250	V_2
r_3	1500	9.0

- Suggest how the intensity of radiation reaching the thermopile varies with the absolute temperature, T , of the sphere.
- Determine the values of r_3 and V_2 . [7 marks]

Total 20 marks

- 9.**
- State FOUR of the basic postulates of the kinetic theory of an ideal gas.
 - Write an expression connecting the pressure of an ideal gas with the mean square speed of the molecules.
 - Using the expression you stated above and the ideal gas equation, $pV = nRT$, derive an expression for the total translational kinetic energy of a monatomic gas. [8 marks]

- Two moles of hydrogen initially at -10°C and $1.0 \times 10^5 \text{ Pa}$, expands at constant pressure to five times its initial volume. The molar mass of hydrogen molecules is $2.0 \times 10^{-3} \text{ kg mol}^{-1}$. Calculate

- the initial volume of the gas
- the final temperature of the gas
- the work done by the gas in the expansion
- the change in the internal energy of the gas during the expansion
- the heat added to the gas
- the root mean square speed, v_{rms} , of the hydrogen molecules after the expansion. [12 marks]

Total 20 marks

END OF TEST